

Title: Customer Churn Analysis

Domain: Telecommunications / Customer Analytics

Presenter: Jay Bourasi

Business Context

Key Objectives:

The goal of this project was to analyze customer behavior, identify key factors influencing churn, and develop data-driven strategies to improve customer retention and reduce revenue loss.

Business Problem:

The company was facing a high customer churn rate due to issues such as short-term contracts, high service charges, and limited customer engagement. This resulted in loss of recurring revenue and increased costs for acquiring new customers.



Summary

Churn Rate:

The overall customer churn rate is **26.58%**, indicating that more than one-fourth of customers leave the company.

Contract Type:

Customers on **month-to-month contracts** show the highest churn compared to one-year and two-year contract customers.

Payment Method:

Customers using **electronic check payments** have the highest churn rate (45.29%), while automatic payment methods show lower churn.

Service Usage:

Customers using **fiber optic internet** and those without services like Tech Support and Online Security are more likely to churn.

Customer Tenure:

New customers with low tenure have a higher likelihood of churning, while long-term customers tend to stay longer.

Pricing Impact:

Customers in higher monthly charge brackets show increased churn, indicating price sensitivity.

Methodology

The project followed a structured approach using various tools:

- ❑ **Data Cleaning (Excel):**

For initial steps include - Removed duplicates, handled null values, and ensured consistency for analysis.

- ❑ **Data Transformation (SQL):**

Extracted, transformed, and aggregated data, generating statistical summaries like averages and trends.

- ❑ **Visualization (Power BI):**

Created dashboards showcasing inventory, procurement, and financial metrics, highlighting trends and distributions.

- ❑ **Presentation (PowerPoint):**

Summarized findings with Power BI visuals and statistical insights for stakeholder communication.

Over All Closure

- ❑ Total Customers: 7,032
- ❑ Lost Customers: 1,869
- ❑ Churn Rate: 26.58%
- ❑ Total Monthly Charges: 455.66K
- ❑ Total Charges: 16.06M
- ❑ Male and female customers are equally distributed
- ❑ High churn in electronic check users
- ❑ Higher charges lead to higher churn
- ❑ Month-to-month contracts show highest churn
- ❑ Fiber optic users have higher churn
- ❑ Low tenure customers are more likely to leave
- ❑ Higher churn without Tech Support
- ❑ Online Security reduces churn
- ❑ Device Protection improves retention

Business Recommendations

- ☐ Promote long-term contracts
- ☐ Encourage auto-payment methods
- ☐ Improve fiber optic service quality
- ☐ Offer loyalty programs
- ☐ Bundle value-added services



Thank You!
For your time and consideration

Best Regards,
Jay Bourasi

